Data Structures and Algorithms · 21CSC201J

Unit 1 · Session 4 · SLO 1 — Solved Worksheet

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Unit â€“ 1 : Session â€“ 4 : SLO â€“ 1 | Self-Referential Structures and Structure Pointers â€” Match the Following

I. Match the Following

1. Self-referential structure â€” P. Has a pointer to the same structure type.
2. Linked list node â€” O. Data + pointer.
3. malloc â€” L. Allocates memory.
4. -> operator â€” R. Access pointer member (ptr->member).
5. . operator â€” M. Access normal structure member.
6. Dynamic node creation â€” T. malloc(sizeof(struct Node)).
7. List head â€” N. First node pointer.
8. Node traversal â€” K. Loop using pointer.
9. Null pointer check â€” J. if (ptr != NULL).
10. Free memory â€” S. free(ptr).
11. Circular list â€” H. Last node points to first.
12. Doubly linked list â€” C. Prev and next pointers.
13. Memory leak â€” E. No free() called.
14. sizeof â€” G. Gets memory size.
15. Data field access â€” B. ptr->data.
16. Insertion at head â€” A. New node before head.
17. List deletion â€” I. Adjust pointers, free memory.
18. List length â€” D. Traverse and count nodes.
19. Node allocation check â€” Q. ptr == NULL (malloc failed).
20. Last node check â€” F. ptr->next == NULL.